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Article Title (110 characters or less)

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OVERVIEW: Provide a 2- to 3-sentence overview of your manuscript. This will be used as a blurb for your article when posting the final publication on our website.

SUMMARY

Visually impaired individuals have to face a lot of problems every day regarding navigation, understanding visual scenes and environments, object localization and reading printed or handwritten text. One of the most frustrating issues that they face is depth perception, as it is directly tied with independent navigation and avoiding hazards. These daily difficulties and limitations of vision put up a barrier to how they interact, how easily they can fetch household objects and how they perceive dynamic and moving bodies. To address these issues, we developed Pixelsense, an AI-powered wearable assistive technology designed to map the depth of surroundings through stereoscopic cameras. We hypothesized that vision can be supplemented using time-synchronized spatial audio and that a diverse set of physical objects can be localized with high accuracy and low latency by utilizing a hybrid architecture of multimodal large language models (LLMs) and vision transformer (ViT) trackers. We conducted multiple distinct experiments to evaluate the effectiveness of the system. We assessed the accuracy of depth estimation through 12 image samples, used 20 trials for testing object localization, and analyzed the effectiveness of the generated spatial audio through 10 human participants. We achieved 95% accuracy in depth estimation, an average of 20px (in 640x480 frames) deviation in object localization and observed that 10 participants correctly identified objects' motion and location solely through auditory feedback. These findings suggest that the approach provides a scalable and accessible model to restore independence for the visually impaired.

INTRODUCTION (needs conclusion!!!)

The estimated number of people affected by visual impairment around the world is 2.2 billion or more (1). Thus, the initiative of creating efficient assistive technologies is critical for public health (1). This immense global burden is not equally shared and disproportionately impacts rural areas, elderly people, and low- and middle-income countries (LMICs), thereby creating major barriers to human development, safe transport, and economic production at the macro-level (1, 2). About 2.5 billion people in the world currently require at least one type of assistive device, but 90% of the people with disabilities in resource-limited regions lack access to these devices (3). Social exclusion and financial exclusion, along with an increased reliance on family support, are the prevalent consequences of the dramatic discrepancy in accessibility (3, 4). Key design factors to consider when designing assistive technologies for blind and

visually impaired (BVI) people include characteristics of the target population and how these users interact with the environment (5).

Traditional white canes have served as the primary tools for navigation for the visually impaired (6). Soon enough, obstacle detection sensors were integrated into these white canes, making them smarter (7). Although they were very easy to use, they came with limitations such as failing to detect hazards within their immediate reach and the inability to provide enough information about the surroundings for the users, which are directly linked to independent mobility (6, 7). Independent mobility and security are vital components in an individual's life; lacking them limits access to social opportunities, work, and schooling in many cases (8). In modern assistive technology (AT) development, the focus has been on the design and development of Electronic Travel Aids (ETAs) to provide visually disabled persons with independent navigation and interaction with their environment (9). One of the most fundamental principles in the engineering science of this kind is sensory substitution - where information collected using one modality (or sense) such as vision is reinterpreted into another (10). However, accurately translating rich and dynamic visual scenes into real-time sensory feedback is still a challenge today, which we are trying to overcome (10). With the advent of deep learning, new methods based on computer vision can perceive interacting elements and obstacles in outdoor and indoor environments in real time for navigation (11).

We hypothesized that vision can be supplemented using time-synchronized spatial audio and that a diverse set of physical objects can be localized with high accuracy and low latency by utilizing a hybrid architecture of multimodal large language models (LLMs) and vision transformer (ViT) trackers. Recent trends in developing navigation technology for the visually disabled no longer focus solely on auditory representation, but also incorporate stereoscopic vision, depth sensing, and three-dimensional spatial audio (12). This can be especially useful for approximately 43.3 million people who are completely blind and 295.1 million people living with moderate-to-severe visual impairment (MSVI) worldwide (13). In order to truly test the effectiveness of our hypothesized solution, we made a full working prototype of a system of a professionally made client-side device and a server. We added two main features for our device, one of which used stereoscopic cameras to generate depth map of the surroundings continuously, then convert it into a 2 second frame of audio for auditory representation. The other feature fused the Gemini 2.5 Flash model, a ViT tracker model, a microphone, and a single camera. The user, trying to find a specific object can just utter its name or describe the object to the microphone, then our vision and audio pipeline will estimate the bounding box for that object and provide directional instruction like a human to the user.

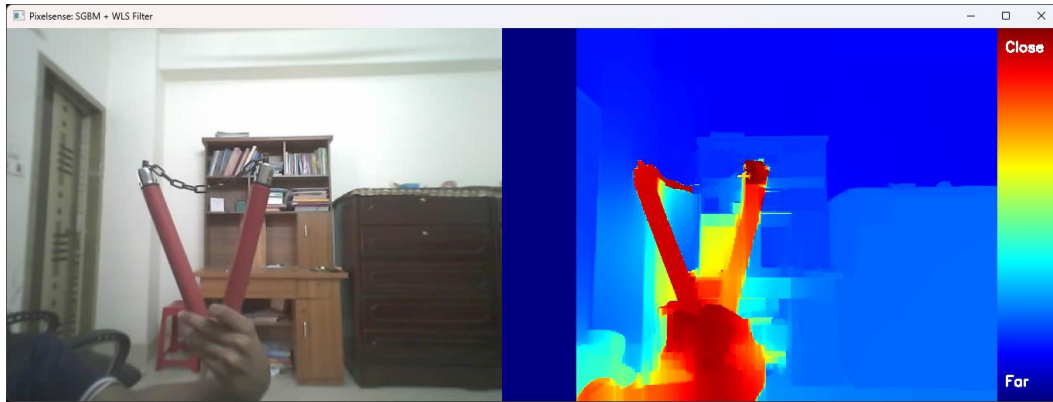
RESULTS

Accuracy of Depth Estimation

The accuracy of depth maps generated from raw image pairs is crucial because reliability of depth estimation directly affects the effectiveness of spatial sound generation. We took **12 image samples** with our stereoscopic system and generated **12 depth maps** corresponding to the images. For each image, we took **3 values** from the depth matrix and compared the value with our measured distance from the camera lens to the physical point using a measuring tape.

$$Accuracy = \frac{Estimated\ depth}{Actual\ depth} \times 100\%$$

The accuracy of each estimated distance was calculated through the **Equation X**. We showed the **3 estimated depth** and their individual accuracy from one sample image (**Figure 1**).



IMPORTANT: ABOVE IMAGE NEEDS TO BE REPLACED WITH RECENT, PROFESSIONAL ONE AND DESCRIPTION NEEDS TO BE ELABORATED. Figure 1. Depth estimation from image. Left image shows the actual image with the actual depths of 3 points. Right image shows the color-coded depth map with estimated distance, error, and accuracy of three target points corresponding to the Left image's 3 points.

Based on the data we collected, it was evident that the accuracy of depth estimation was very high for short ranges (typically 94% accuracy for 5cm to 150cm) and slowly dropped as distance increased (Figure 2). Any accuracy over 80% was perfectly usable and reliable for spatial audio generation, suggesting that our system was reliable for 5cm to 300cm range (Figure 2).

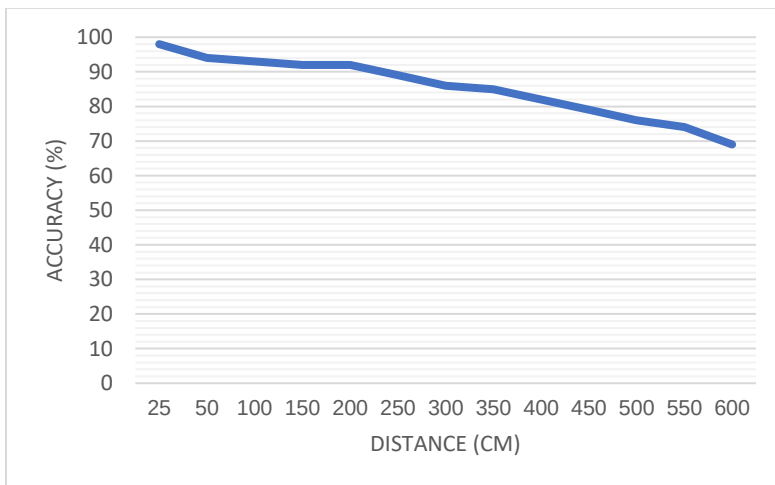


Figure 2. Estimation accuracy vs distance graph. The graph is based on the 36 depth samples from a wide range of distance (25cm to 600cm). Each point on the line represents the expected accuracy (y) for measuring a distance (x). The very small negative slope of the line indicates negligible accuracy drop with increased distance.

Effectiveness Spatial Audio Generation

After each depth map is generated, they are converted into stereo audio for the user. To transform the depth map into audible sound, X, Y, and Z axes of the visual field is correlated to time/panning, pitch and volume respectively (24). To determine the effectiveness of the audio representation of the depth map for navigation and environmental understanding, we let our device be used by 10 visually impaired participants. Each of them, wearing our device was assessed through three tests: determining an object's position, determining an object's motion and safe navigation.

Except for the safe navigation test, the other two tests were taken with a red ball as an object. In safe navigation test, the participants were taken to an unknown room, and were told to get out of the room solely depending on the audio feedback. We showed the test results of all participants in **(Figure 3)**.

Participant ID	Age	Sex	Test Results			
			Trial No.	OPD (second)	OMD (second)	SN (second)
P1	22	M	1st	1.5	1.8	12
			2nd	2.0	1.5	
			3rd	1.7	1.7	
			Mean	1.73	1.6	
P2	22	M	1st	1.5	1.8	12
			2nd	2.0	1.5	
			3rd	1.7	1.7	
			Mean	1.73	1.67	
P3	22	M	1st	1.5	1.8	12
			2nd	2.0	1.5	
			3rd	1.7	1.7	
			Mean	1.73	1.67	
P4	22	M	1st	1.5	1.8	12
			2nd	2.0	1.5	
			3rd	1.7	1.7	
			Mean	1.73	1.67	
P5	22	M	1st	1.5	1.8	12
			2nd	2.0	1.5	
			3rd	1.7	1.7	
			Mean	1.73	1.67	
P6	22	M	1st	1.5	1.8	12
			2nd	2.0	1.5	
			3rd	1.7	1.7	
			Mean	1.73	1.67	
P7	22	M	1st	1.5	1.8	12
			2nd	2.0	1.5	
			3rd	1.7	1.7	
			Mean	1.73	1.67	
P8	22	M	1st	1.5	1.8	12
			2nd	2.0	1.5	
			3rd	1.7	1.7	
			Mean	1.73	1.67	
P9	22	M	1st	1.5	1.8	12
			2nd	2.0	1.5	
			3rd	1.7	1.7	
			Mean	1.73	1.67	
P10	22	M	1st	1.5	1.8	12
			2nd	2.0	1.5	
			3rd	1.7	1.7	
			Mean	1.73	1.67	
Mean				1.84	2.1	14

OPD – Object Position Determination, OMD – Object Motion Determination, SN – Safe Navigation

Figure 3. Test result of participants. Participants successfully pointed towards the position of a red ball held by another person's hand in the air in all three trials with a mean response time of **1.84s**. Participants also uttered the motion of a red ball in the air (moved by another person) verbally ('left', 'right', 'up' etc.) within a mean time of **2.1s** after movement. The participants also successfully got out of a room within **14s**. All were done solely through stereo spatial audio cues.

DISCUSSION

Our results supported that continuous depth map estimation of surroundings with high accuracy (> 90%) using cheap camera modules and embedded systems is indeed possible and reliable by offloading computationally expensive tasks to a server (Figure 2). The survey and tests on 10 visually impaired participants exhibited positive results as the participants could determine position and motion of objects solely through spatial audio cues (mean response time of 1.84s and 2.1s for object position determination and object motion determination tests respectively) generated by our device. Moreover, the safe navigation (SN) test showed that participants were able to navigate with minimal effort indoors.

MATERIALS AND METHODS

Hardware Configuration and Integration

The physical device, which is a piece of technology attached to a spectacle frame, is one of the core parts of Pixelsense. This device contains a 3D-printed (printed using PLA filament) frame which houses the main circuit board and electronic components. The two OV3660 3-megapixel cameras are placed on top of the frame housing, and the base to which they are attached is kept as flat as possible to maintain identical vertical and forward alignment for both sensors. The two cameras are placed on the same horizontal axis with their optical centers 59-mm apart.

- [
- 2 images of the device:
1. The full, intact device with frame
 2. Only the front and back side of the circuit board

]

The device contains two independent but connected systems, which are centered around the Seeed Studio XIAO ESP32 S3 Sense (14). Each of these boards contains the necessary ESP32-S3 microcontroller unit (MCU) and a Flexible Printed Circuit (FPC) connector (to which we connect the cameras via an FPC cable) and features an integrated digital Micro-Electro-Mechanical Systems (MEMS) microphone. We soldered the two ESP32 boards onto a custom-designed Printed Circuit Board (PCB).

A MAX98357A (15) chip-based board, which integrates a digital to analog converter (DAC) and a mono audio amplifier, is connected to each of the ESP32-S3 board via Inter-Integrated Sound (I2S) protocol. These take in digital audio samples, modulate the samples into analog signals, amplify them, and play them via the earphones.

A single VL53L0X chip-based time of flight (TOF) infrared-laser distance sensor (16), which communicates with the ESP32-S3 via Inter-Integrated Circuit (I2C) protocol, is soldered onto the PCB. This sensor is used to quickly detect any obstacles within 1-meter range.

We have also added two TTP223 chip-based capacitive touch sensors for the user to activate different features and modes of the device (17). A 22pF capacitor is soldered to each of the touch sensors for better touch sensitivity.

The two ESP32-S3 MCUs communicate with each other via Universal Asynchronous Receiver-Transmitter (UART) protocol, which allows these two independent systems to send synchronous image streams (captured by the cameras) to the server and receive audio samples from the server to play them through the stereo speakers synchronously. Although we have two microphones in the device (one in each ESP32-S3 board), we only utilize one of them to capture user inputs.

Server Configuration and Integration

The server, for our testing is a consumer laptop. The server handles all the complex and computationally expensive tasks like rectifying stereoscopic images, generating pixel disparity and depth map, generating stereo audio samples, speech recognition, text to speech generation, running transformer models, and calling external APIs. The program within the server that handles everything is written in Python 3.10.

It utilizes Transmission Control Protocol (TCP) to establish a connection with the ESP32-S3. Whenever a user touches a touch sensor, the activation command associated with the gesture is sent to the server via TCP. Whenever the device turns on, the two ESP32-S3 individually establish the connection with the server. This way, the server knows which feature to activate. Moreover, User Datagram Protocol (UDP) is utilized to send and receive image and audio sample streams with minimal network overhead. Here, websockets 15.0.1 (18) library was used for handling TCP connection and socket library (19) was used to handle UDP connection.

Depth Map Estimation and Spatial Sound Generation

This feature allows a visually impaired individual to perceive the objects and their motion through melodic frequencies of sound. To activate this feature, the user presses a touch sensor. The ESP32 detects it via its digital pin and sends a message string to the server for the initialization of the feature. The two ESP32 boards coordinate with each other via UART and stream images of the user's surroundings to the server synchronously, making sure every pair of image frame captured by the cameras are taken at the same time. The server, receiving the image streams first rectifies the pair of images with the OpenCV's stereoRectify method (20). To rectify the image pairs, first we obtained the intrinsic and extrinsic parameters of the camera by the chessboard calibration method with 40 image samples taken at diverse distance and angles for better accuracy and to minimize stereo reprojection error (21). According to the pinhole camera model, a point in 3D space, $P = [X, Y, Z]^T$ is projected onto a 2D plane at pixel coordinates $[u, v]^T$. We use the intrinsic camera matrix (K) to represent this relation:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \quad (\text{Equation 1})$$

In Equation 1, s is a scaling factor (representing depth). The intrinsic camera matrix K for each of the cameras has the following form, which contains the focal lengths in pixels (f_x, f_y) and principal points (c_x, c_y):

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{Equation 2})$$

Cameras, especially the OV3660 camera which contains a plastic lens is prone to radial and tangential distortion. To solve this, we obtained the distortion coefficient vector $D = [k_1, k_2, p_1, p_2, k_3]$ during camera calibration, where k_1, k_2 and k_3 are radial distortion coefficients and p_1 and p_2 are tangential distortion coefficients. These intrinsic parameters (camera matrix and distortion coefficient vector) are obtained through OpenCV's calibrateCamera method for each of the two cameras.

Let P_l and P_r are two points viewed by the left and right cameras respectively, which corresponds to the same 3D point in the scene. The geometric transformation between the optical centers of two cameras is represented by a 3x3 rotation matrix (R) and a 3x1 translational vector (T), representing the physical distance between the two cameras' optical centers:

$$P_r = R \cdot P_l + T \quad (\text{Equation 3})$$

By utilizing OpenCV's stereoCalibrate method, we solve for the rotation matrix and translational vector, which are the extrinsic parameters. We stored the extrinsic and intrinsic matrices as a JavaScript Object Notation (JSON) file in our server after calibration.

By feeding these extrinsic and intrinsic matrices to `initUndistortRectifyMap` method of OpenCV, and remapping the images, a 3D point P was projected onto the left and right rectified images such that the vertical coordinate is identical. During image stream, every single image is rectified and then goes through multiple algorithms and pipelines of disparity estimation, post-filtering optimization, and geometric transformation. Every pair of images are converted into grayscale and Semi-Global Block Matching (SGBM) algorithm is applied (22). Through this algorithm, two raw pixel disparity maps are generated for the left and right cameras. To enhance smoothness of the output, a Weighted Least Squares (WLS) fast-global filter is then applied to the raw left disparity map, using the left grayscale image as a structural guide to refine boundaries (23). According to our tests, a filtering parameter $\lambda = 8000$ and a color sensitivity parameter $\sigma_{color} = 1.5$ worked the best. The resulting filtered sub-pixel disparity map is downsampled by a factor of 16.0 to reconstruct the true pixel disparity (d). The absolute pixel distance Z (in millimeters) of each pixel coordinate is calculated using triangulation across similar triangles. The absolute depth map is generated by applying the simple formula for each value from the pixel disparity map:

$$Z = \frac{f \cdot B}{d} \quad (\text{Equation 4})$$

Here, f is the focal length of the camera sensor in pixels, extracted from the camera matrix element $K[0,0]$. B is the baseline distance between the optical centers of the two cameras derived from the absolute horizontal translation vector component $|T_x|$. d represents the true pixel disparity computed via the WLS filter.

To transform the depth map into audible sound, X, Y, and Z axes of the visual field is correlated to time/panning, pitch and volume respectively (24).

[Show a depth map with x, y, z labeled with pitch, time and volume]

To represent vertical position of an object, the vertical axis is divided into 20 horizontal bands and each band is mapped to a unique audio frequency. To provide pleasant listening experience, a Major Pentatonic Scale is used which begins at a base frequency of 130Hz and is capped at 3kHz to prevent audio distortion. Higher frequency corresponds upper parts of the depth map, while lower frequencies indicate lower objects. The horizontal field of view is represented by “scanning” the image (depth map) from left to right over a definite period. The time of one frame scanning is flexible and can be set by the user. We use a 2 second time frame for testing. As the scan progresses from left to right, the audio signal is distributed between the left and right stereo speakers using an Equal Power Panner (Equation 5) keep the panning natural, it follows an S-curve function:

$$S(p) = \frac{1 - \cos(\pi \cdot p)}{2}$$

Here, p represents the progress of panning in 0 to 1 range.

$$A_L(p) = \sqrt{1 - S(p)} \quad (\text{Equation 6})$$

$$A_R(p) = \sqrt{S(p)} \quad (\text{Equation 7})$$

The amplitude for the left and right channel is given by the functions $A_L(p)$ and $A_R(p)$ respectively.

The proximity of the objects (Z) is directly represented by the amplitude of the audio signal. At first the values of depth map are normalized against a predefined maximum distance (Z_{max}) to get a range of 0 to 1. The normalized depth values are raised to the power of two. This ensures objects close to the user sound louder compared to insignificant background noise.

Determining Accuracy of Depth Estimation

The accuracy of depth map was calculated by comparing the actual depths (distance from the camera lenses) and the measured depth at different points of sample scenes. We kept the device stable on a stand. Before capturing an image,

we determined the distance from the camera lens to some objects, corners, and points of the scene by using a measuring tape. This way, we got the actual depths of some points of a scene. After that, we captured an image of that scene with the device and obtained the depth map matrix. By utilizing a python script, we created a graphical user interface which allowed us to easily fetch the estimated depth of every point of the image from the depth map.

$$Accuracy = \frac{Estimated\ depth \times 100\%}{Actual\ depth}$$

The accuracy of measurement was determined through the Equation 8. In this way, we determined the accuracy of depth estimation for a total of **36 points from 12 image samples**. The average accuracy turned out to be **94%**. The range of distance measured and estimated during accuracy determination was 10 cm to 300 cm.

Determining Effectiveness of Spatial Audio

The effectiveness of spatial audio was determined by observing how users avoided hazards and perceived the motion and position of objects. We let our device be used by **10 visually impaired individuals**. The people were given one hour to get familiar with the spatial audio cues. After that, 3 tests were given to the participants. The first one was to determine the position of a red ball by pointing towards it. It was in midair held by a person's hand. In the second test, a person moved a red ball in the air in specific directions. The participant had to tell the direction of movement by describing it vocally. In these two tests, we noted down how fast the participant responded in a notepad. In the final test, the participant was taken to a room, from which he had to escape solely depending on audio cues generated by our device. We noted down the time taken for the participant to escape.

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354 **Appendix**